Hierarchical Biometrical Genetic Analysis of Longitudinal Dynamics

Kevin L. McKee
Virginia Tech
Center for Biostatistics and Health Data Science

Abstract

For many phenotypes, individual scores are obtained as the parameter estimates of person-level models fit to intensive repeated measures from physiological sensors or experience sampling studies. Biometrical genetic analysis of such phenotypes is often done in a 2-step sequence: first the phenotypic parameters are estimated for each individual, then classical twin modeling is used to partition their variance. This study demonstrates deficiencies in accuracy and statistical power of the two-step approach to estimate genetic signals and advocates for the use of hierarchical models to overcome both problems. Simulations are used to demonstrate the benefits to accuracy and statistical power from a hierarchical modeling approach. A model of heart rate fluctuations was applied to experimental data from twin pairs recorded in independent trials. Results of the data application reveal moderate but uncorrelated heritabilities for two parameters of heart rate: oscillation frequency and damping ratio. By merging biometrical genetic analysis with process models, hierarchical mixed-effects modeling has potential to assist with discovery and extraction of novel phenotypes from within-person data and to validate theoretical models of within-person processes.

Introduction

A principal goal of behavioral genetics (BG) is to make nomothetic inferences about the nature of inter-individual variation in particular phenotypes (Plomin, 1990). This is accomplished in most cases with cross-sectional or panel designs that measure the phenotype over one or few occasions. However, many phenotypes cannot be observed or modeled without taking numerous repeated measures to reveal temporal patterns. For example, Ecological Momentary Assessment (EMA, Shiffman et al., 2008) or Experience Sampling (Hektner, 2007) studies collect hourly or daily subjective reports to assess variability in affect for trait-like stability and association with personality or other complex traits (Röcke et al., 2009; Eid and Diener, 1999; Kuppens et al., 2007). In many experiments, phenotypes are defined by sampling physiological states at high frequency, such as using electrocardiogram (ECG) to observe variability in heart rate (Rajendra Acharya et al., 2006), or using electroencephalography (EEG) and functional magnetic resonance imaging (fMRI) to observe

functional connectivity of brain regions (Bastos and Schoffelen, 2016). In such examples, the hierarchical data sets produced by these designs differ from simpler longitudinal panels primarily in the number of repeated measures involved, which may be hundreds or thousands, with measurement intervals ranging from milliseconds to days. The standard array of longitudinal structural equation models (see: SEM, Bollen, 1989) common to Behavior Genetics, such as latent growth curves (Mcardle and Epstein, 1987), simplex models (eg. Dolan et al., 2014), and cross-lagged panels (Moscati et al., 2018), are not computationally suited for modeling more than a small number of occasions per person and are limited to only the simplest kinds of time-dependence, such as polynomial trends and first-order autoregressions.

The complexity of analyzing intensive longitudinal data across many individuals, i.e., time series, is simplified if within-person processes can be summarized by indices of temporally stable traits, known in quantitative psychology as *dynamics* (Vallacher et al., 2015; Boker, 2002; Nesselroade, 2002). Specifically, dynamics may refer to model parameters that characterize time dependence, changes of state, or patterns of motion in the data. A popular model of emotion regulation serves as one of the simplest examples, defining regulation as the rate of decay in affect toward an equilibrium or mean value (Oravecz et al., 2011; McKee et al., 2020). The model is specified according to the theory that variation in affect ratings over time reflects the accumulation of emotional perturbations due to random, unmeasured environmental stimuli. Latent, intrinsic mechanisms of regulation thus determine the rate of accumulation and can be estimated from the rate and extent of variability in each individual's series of ratings.

With high-resolution sensor data, more complex time series models can potentially index a variety of physiological mechanisms from subtle patterns of variation in even a single, univariate series. One example from previous work involved a theoretical model of human balance to obtain indices of neurological, cognitive, and muscular functioning from sensor data sampled one hundred times per second (McKee and Neale, 2019). Results were informative: Some estimates confirmed the distributions of indices that had only been previously conjectured, while others replicated those that had otherwise required specialized measurement instruments. If the same techniques were combined with biometrical genetic modeling, inter-individual variance in such indices could be partitioned into genetic and environmental sources. Historically, heritabilities have been used as comparative evidence for the biological basis of phenotypes (Eysenck and Prell, 1951). Similarly, the heritabilities of dynamics estimated under competing theories may be weighed in their comparison and used to guide the discovery of endophenotypes such as those used to characterize postural control.

In terms of statistical methods, a biometrical genetic analysis of dynamics requires first obtaining individual estimates of the phenotypic model parameters, then decomposing the distribution of those estimates into genetic and environmental sources of variation. There are consequential drawbacks to conducting each stage of the analysis separately. First, two-step analyses are statistically inefficient because between-persons information is not considered during the estimation of within-person parameters and vice versa. Second, it is difficult to accurately integrate sampling variance through each level of the analysis, and neglecting to do so potentially invalidates tests and inferences that rely on accurate knowledge of parameter sampling error. These issues are resolved by using a unified hierarchical

model in which the two-way dependence between dynamics and their variance components factors into the estimation of both (e.g., McElreath, 2016, Ch.13). To date, complete hierarchical models for the biometrical genetic analysis of individual-level parameters have not been widely used or studied for that purpose, and existing methods of estimation are not well known.

This study provides a general guide to hierarchical state-space modeling for biometrical genetic studies with comparisons to analogous two-step analyses. In the first section, state-space models are introduced for modeling time series data. In the second section, simulations are used to determine statistical biases and power for two step biometrical genetic analysis of state-space dynamics, with comparisons to hierarchical modeling. The final study involves an application of hierarchical state-space modeling to experimental heart rate data.

Mixed-Effects State-Space Models for Twin Studies

Within-person data are not always time-indexed, but usually multiple measurements of single type must be taken in sequence such that chronology is relevant to the model. This study thus focuses on the case of time series models. State-space models (SSM) are a general class of models that subsumes Markov processes and SEM. In an SSM, latent variables are often called *latent states* or just *states* and are specified to vary over time according to Markov processes in either discrete or continuous time. The states are inferred from a measurement model that incorporates one or more noisy indicators as in a factor analysis.

Continuous-time SSMs have the form of stochastic differential equations (SDEs), which are convenient for many reasons. Parameter values of an SDE are invariant to measurement intervals and can provide parsimonious representations of complex dynamical systems. For more detailed introductions, resources include Wei (2006); Douc (2014); Åström and Murray (2008); Bar-Shalom et al. (2002). An introduction to SSMs with implementation in `R`(R Core Team, 2019) SEM modeling package `OpenMx` (Neale et al., 2016) is given by Hunter (2018). Nonlinear SSMs are implemented in the `dynr` package (Ou et al., 2019). Broader implementation of SSMs with hierarchical modeling are available in `R` package `ctsem` (Driver et al., 2017) and more recently in Mplus (McNeish and Hamaker, 2020).

For complex trait genetics we are concerned with the sources of inter-individual variation in phenotypic scores. If a phenotype is operationalized as the dynamics of an SSM, then we must include random effects in the SSM to estimate both the individual dynamics and their distribution. A biometrical genetic mixed-effects model assigns a probability distribution to data y conditional on both phenotypic parameters (θ) and biometrical genetic parameters that define their twin covariances, which I will denote ACE[2].

$$P(\mathbf{y}|\boldsymbol{\theta}, ACE) = \frac{P(\boldsymbol{\theta}, ACE|\mathbf{y})P(\mathbf{y})}{P(\boldsymbol{\theta}, ACE)} = \frac{P(\boldsymbol{\theta}|ACE, \mathbf{y})P(ACE|\mathbf{y})P(\mathbf{y})}{P(\boldsymbol{\theta}|ACE)P(ACE)} \tag{1}$$

[2] ACE refers to a twin model with additive genetic (a), common environmental (c), and unique environmental (e) variance components but is used colloquially to symbolize twin modeling in general, though other specifications such as allelic dominance (d) are common.

This section will describe three main components of the model: the state equations in either discrete time as autoregressions or continuous time as SDEs, followed by incorporation of the measurement model by which states are inferred from the data, then the hierarchical biometical genetic variance components. Example model matrices will be constructed along the way then brought together in the likelihood function.

Discrete time. A state-space model in discrete time can be stated generally as,

$$\mathbf{x}_t = \mathbf{g}[t, \mathbf{x}_{t-1}, \mathbf{u}_{t-1}, \mathbf{q}_{t-1}, \boldsymbol{\theta}_x] \tag{2}$$
$$\mathbf{y}_t = \mathbf{h}[t, \mathbf{x}_t, \mathbf{u}_t, \mathbf{r}_t, \boldsymbol{\theta}_y] \tag{3}$$

Linear state-space models, i.e, those in which observations (y) are proportionally related to latent states (x), and states are proportionally related to previous states (x_{t-1}), can be given in matrix form. Each matrix is subscripted by individual or family i to indicate that we will fit many such models in a twin study:

$$\mathbf{x}_{i,t} = \mathbf{A}_i\mathbf{x}_{i,t-1} + \mathbf{B}\mathbf{u}_{i,t} + \mathbf{q}_{i,t}, \quad \mathbf{q}_{i,t} \sim \mathcal{MVN}(0, \mathbf{Q}_i), \tag{4}$$
$$\mathbf{y}_{i,t} = \mathbf{C}\mathbf{x}_{i,t} + \mathbf{D}\mathbf{u}_{i,t} + \mathbf{r}_{i,t}, \quad \mathbf{r}_{i,t} \sim \mathcal{N}(0, \mathbf{R}), \tag{5}$$
$$t \geq 0, \quad i \in \{1...N\}. \tag{6}$$

The $\mathbf{A}$ matrix contains the auto and cross-regressive coefficients of the latent state, $\mathbf{x}_t$. $\mathbf{B}$ contains the regression coefficients on potentially time-varying covariates, or external forcing functions $\mathbf{u}_t$. $\mathbf{q}_t$ is a noise variable representing random disturbances to the state, most often taken to be normally distributed. The $\mathbf{C}$ matrix contains the factor loadings, or coefficients that relate the indicators to the latent state. $\mathbf{D}$, similarly to $\mathbf{B}$, includes regression coefficients of $\mathbf{y}_t$ onto time-varying covariates. Finally, $\mathbf{r}_t$ is a Gaussian distributed white noise variable that represents measurement error, or additional random variation that does not influence the latent state. When the $\mathbf{A}$ matrix contains only zeros, the model reduces to a factor model.

For example, take x to be a lag-1 vector autoregression (VAR) with two latent states $\mathbf{x}_t$ sharing a single time-varying covariate $\mathbf{u}_t$. The states and parameters of the model will be particular to family or individual i.

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{i,t} = \begin{bmatrix} \beta_1 & \gamma_{1,2} \\ \gamma_{2,1} & \beta_2 \end{bmatrix}_i \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{i,t-1} + \begin{bmatrix} \alpha \\ \alpha \end{bmatrix} u_{i,t} + \begin{bmatrix} q_1 \\ q_2 \end{bmatrix}_{i,t}, \tag{7}$$
$$\mathbf{q}_t \sim \mathcal{MVN}([0,0], \sigma_{q_i} \begin{bmatrix} 1 & \rho_i \\ \rho_i & 1 \end{bmatrix}), \tag{8}$$

This example model could be used either to represent two dynamically linked (i.e., state dependent) behaviors within an individual or to represent a single behavior measured in dynamically linked twins. The latter usage is suitable for experience sampling studies of twin pairs that interact, cohabit, or in any way share their environments during the study period. In that case, parameters γ and ρ characterize cross-twin influence and shared disturbances.

Continuous time. Continuous time state-space models use SDEs to specify change in the latent state $\dot{x}(t)$ as a function of the level $x(t)$. Other components remain virtually the same as discrete time models, except that $t \in \mathbb{R}$, so $x(t)$ refers to the solution of the differential equation at time t.

$$\dot{\mathbf{x}}(t) = \mathbf{g}[t, \mathbf{x}(t), \mathbf{u}(t), \mathbf{q}(t), \boldsymbol{\theta}_x] \tag{9}$$
$$\mathbf{y}(t) = \mathbf{h}[t, \mathbf{x}(t), \mathbf{u}(t), \mathbf{r}(t), \boldsymbol{\theta}_y] \tag{10}$$

For linear continuous time models, the state equation is given as a matrix SDE:

$$\dot{\mathbf{x}}_i(t) = \mathbf{A}_i\mathbf{x}_i(t) + \mathbf{B}\mathbf{u}(t) + \mathbf{q}_i(t). \tag{11}$$

A convenient way to fit differential equation models is to discretize matrices $\mathbf{A}$, $\mathbf{B}$, and $\mathbf{Q}$ according to time interval Δ., i.e., solve the SDE for each new successive measurement interval. The discretized matrices can then be used much like the discrete time model:

$$\mathbf{x}_i(t) = \mathbf{A^d}_i\mathbf{x}_i(t - \Delta) + \mathbf{B^d}\mathbf{u}(t) + \mathbf{q}_i(t). \tag{12}$$

Measurement models. In most applications, the time dependence of the observed data $\mathbf{y}$ is modeled by way of the latent state $\mathbf{x}$. A measurement model is thus used to regress one or more observed indicators on the latent state and account for additional independent and identically distributed noise or model residuals particular to each. Whereas our state equation contains the individual or family-level phenotypic parameters, the factor loadings and residuals may be particular to the instrument, i.e., questionnaire or physiological sensors, and are thus shared across groups.

We can extend our example by supposing x_1 and x_2 represent twins responding on the same k questionnaire items, with each item subject regressed on a common time-varying covariate u and having its own residuals r:

$$\begin{bmatrix} y_{1,1} \\ \vdots \\ y_{1,k} \\ y_{2,1} \\ \vdots \\ y_{2,k} \end{bmatrix}_{i,t} = \begin{bmatrix} \lambda_1 & 0 \\ \vdots & 0 \\ \lambda_2 & 0 \\ 0 & \lambda_1 \\ 0 & \vdots \\ 0 & \lambda_2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}_{i,t} + \begin{bmatrix} \gamma_1 \\ \vdots \\ \gamma_k \\ \gamma_1 \\ \vdots \\ \gamma_k \end{bmatrix} u_{i,t} + \begin{bmatrix} r_{1,1} \\ \vdots \\ r_{1,k} \\ r_{2,1} \\ \vdots \\ r_{2,k} \end{bmatrix}_{i,t}, \tag{13}$$

$$r_{1,k}, r_{2,k} \sim \mathcal{N}(0, \sigma_{r_k}). \tag{14}$$

Hierarchical genetic models. Any parameters of the state and measurement equations may be person or family-level, but our hypotheses likely only concern the genetic variation in those with particular theoretical implications. Individual dynamics of the state-space model, indexed by family i and twin j, are distributed according to a biometrical genetic model-implied covariance matrix, $\Sigma_\theta(ACE)$:

$$\boldsymbol{\theta_{i,j}} \sim \mathcal{MVN}(\boldsymbol{\mu_\theta}, \Sigma_\theta(ACE)) \tag{15}$$

If, in our example, $\mathbf{y}$ were to represent reported states of affect, then the autoregressive coefficients of the state, β_1 and β_2, would represent the homeostatic tendency specific to each twin. For a model-implied covariance matrix of β, take the univariate ACE model:

$$\Sigma_\theta(ACE) = \sigma_\theta \begin{bmatrix} a^2 + c^2 + e^2 & z_i a^2 + c^2 \\ z_i a^2 + c^2 & a^2 + c^2 + e^2 \end{bmatrix}, \quad z \in \{0.5, 1\}. \tag{16}$$

In this notation, z is the coefficient of average genetic relatedness or zygosity, which in monozygotic twins is 1.0 and in dizygotic twins is 0.5. The parameters a, c, and e represent the factor loadings of the phenotype onto latent variance components representing additive genetic, common environmental (e.g., familial), and unique environmental sources of variance. a^2 is also known as the heritability of the phenotype. In principle, many genetic models are possible. Whereas this is the simplest univariate specification, multivariate twin modeling techniques will be demonstrated later to study the genetic covariances among multiple phenotypes estimated from the same model.

Data structure. Data structures for mixed-effect SSMs models may be complex, and the ideal structure depends on what is expected by the chosen software. High-dimensional arrays (matrices with more than two indices) provide a compact representation. For instance, data may be subscripted in a top-down, hierarchical order, $\mathbf{y}_{i,j,k,t}$, starting family index i, twin index j, variable index k, and time index t. Each zygosity is then given its own array to allow separate sample sizes. Supplemental Stan code for this study are provided that use this format. Alternatively, the data may be specified in "long form" as $\mathbf{y}_{t,k}$ with rows representing occasions of measurement, and columns representing variables. Zygosity, family, and twin indices are then included in additional columns and the data are separated out within the model software. For continuous-time models, it is essential to included a column of times for each measurement occasion to correctly account for varying measurement intervals.

Estimation. To fit the model, the hierarchical likelihood function is formulated according to

$$P(\theta, ACE|y) = \frac{P(\mathbf{y}|\theta, ACE)P(\theta|ACE)P(ACE)}{P(\mathbf{y})}. \tag{17}$$

If the model is fit in a frequentist context, then the denominator $P(\mathbf{y})$ (a constant) and the prior probability distribution $P(ACE)$ are excluded, leaving $P(\theta, ACE|y) = P(\mathbf{y}|\theta, ACE)P(\theta|ACE))$.

Specific representation of the likelihood function of this model depends on the estimation strategy. SSMs may be implemented with the Kalman-Bucy filter (Kalman and Bucy, 1961; Kalman et al., 1960). In brief, the Kalman-Bucy filter is an algorithm for producing minimum-variance unbiased estimates of latent states conditional on the observed data when both process disturbances $\mathbf{q}$ and measurement residuals $\mathbf{r}$ are normally distributed. The algorithm recursively predicts each new state ($\hat{\mathbf{x}}_{i,t}$) from the preceding observations, decomposes the prediction error into measurement and process noise distributions with covariances $\hat{\mathbf{R}}$ and $\hat{\mathbf{Q}}$, respectively, then updates the state estimate using the most recent data. The likelihood therefore involves only the total prediction error, ($\mathbf{y}_{i,t} - \mathbf{C}\hat{\mathbf{x}}_{i,t}$) and the total variance-covariance matrix ($\hat{\mathbf{S}}$) of the observations. These are computed for every observation within-person, and each person-level likelihood includes the likelihood of that person's

phenotypic parameters (θ_i) given their biometrical genetic model-implied covariance.

$$P(\theta, ACE|y) = \prod_{i=1}^{N} \left[\prod_{t=0}^{T} \left[\frac{\exp\left(-0.5(\mathbf{y}_{i,t} - \mathbf{C}\hat{\mathbf{x}}_{i,t})\hat{\mathbf{S}}_{i,t}^{-1}(\mathbf{y}_{i,t} - \mathbf{C}\hat{\mathbf{x}}_{i,t})^{\mathsf{T}}\right)}{\sqrt{(2\pi)^k|\hat{\mathbf{S}}_{i,t}|}} \right] \times \frac{\exp\left(-0.5(\theta_i - \mu_\theta)\hat{\mathbf{\Sigma}}_\theta(ACE)^{-1}(\theta_i - \mu_\theta)^{\mathsf{T}}\right)}{\sqrt{(2\pi)^k|\hat{\mathbf{\Sigma}}_\theta(ACE)|}} \right] \tag{18}$$

Alternatively, SSMs can be fit to data without the Kalman filter when using Markov Chain Monte Carlo (MCMC). In that case, samples of the latent state are drawn from their posterior distribution along with those of the phenotypic parameters. In that case, the distributions of $\mathbf{q}$ and $\mathbf{r}$ have separate terms:

$$P(\theta, ACE|y) = \prod_{i=1}^{N} \left[\prod_{t=0}^{T} \left[\frac{\exp\left(-0.5(\mathbf{x}_{i,t} - \mathbf{A}\mathbf{x}_{i,t-1})\hat{\mathbf{Q}}_i^{-1}(\mathbf{x}_{i,t} - \mathbf{A}\mathbf{x}_{i,t-1})^{\mathsf{T}}\right)}{\sqrt{(2\pi)^k|\hat{\mathbf{Q}}_i|}} \times \frac{\exp\left(-0.5(\mathbf{y}_{i,t} - \mathbf{C}\mathbf{x}_{i,t})\hat{\mathbf{R}}^{-1}(\mathbf{y}_{i,t} - \mathbf{C}\mathbf{x}_{i,t})^{\mathsf{T}}\right)}{\sqrt{(2\pi)^k|\hat{\mathbf{R}}|}} \right] \times \frac{\exp\left(-0.5(\theta_i - \mu_\theta)\hat{\mathbf{\Sigma}}_\theta(ACE)^{-1}(\theta_i - \mu_\theta)^{\mathsf{T}}\right)}{\sqrt{(2\pi)^j|\hat{\mathbf{\Sigma}}_\theta(ACE)|}} \right] \tag{19}$$

By this method, the model is not restricted to multivariate normality, and in principle any distributions can be used.

Hierarchical models such as this are not generally fit with gradient descent due to mathematical and computational challenges. For any model of the above form, there exists a singularity in which $\sigma_\theta = 0$, all $\theta_i = \mu_\theta$, and thus $P(\theta|ACE) = \infty$. Algorithms to maximize the likelihood will be drawn directly to that singularity. The problem is mitigated if there are enough occasions of measurement per person such that local solutions of θ are sufficiently convex (i.e., precise) to prevent the optimizer from straying toward the singularity.

With MCMC, the parameter vector does not follow the gradient of the likelihood function directly to the posterior maximum, i.e., the maximum likelihood estimate (MLE). Rather, MCMC probabilistically explores the likelihood function such that the trace of its path approximates the sampling distribution of the parameters. As the number of random effects parameters increases, the probability of the parameter vector stepping directly toward the MLE becomes vanishingly small. For this reason, both the MLE and the singularity are avoided when the number of random effects is large, and instead, the area surrounding the MLE, known as the *typical set*, is populated (For *typicality*, see Ch. 4.4, MacKay, 2003). The expected values of the typical set give reasonable point estimates, particularly in random-effects models where the likelihood includes a singularity. Approximate marginal MLEs may be obtained however by taking the multivariate maximum *a posteriori* (MAP) over a small subset of the parameters. This is useful for our purposes,

where the parameters a, c, and e are of central interest but have highly correlated sampling distributions.

The analyses to follow used MCMC both to examine the properties of models under small samples and to visualize the sampling distributions of parameters. Low-dimensional MAPs were obtained using multivariate Gaussian density kernel estimation. Sampling for the subsequent analyses was performed using the No-U-Turn Sampler (Hoffman and Gelman, 2014) as part of modeling software Stan (Carpenter et al., 2017).

To improve convergence, weakly-regularizing prior distributions were chosen for all parameters with variances at least two orders of magnitude larger than the expected variances of each parameter. Any biases that result from their use are expected to be negligible. As a precaution, priors distributions for a, c, and e were in all cases specified with means at zero for a and c components, and means at one for e components, such that any bias incurred errs toward the null hypothesis of no genetic signal.

Simulation: 2-Step Estimation versus Hierarchical Modeling

A simulation was designed to compare heritability estimation for dynamics between two approaches. In the first, maximum likelihood was first used to estimate twin-level autoregressive parameters, then in a second, separate step, the heritability of those parameters was also estimated by maximum likelihood. In the second approach, a hierarchical random-effects autoregression model was specified that was identical to the previous, except all individual level parameters and their heritability were estimated jointly with MCMC.

The comparison of estimation strategies was simplified to first-principles by taking the simplest relevant model: a mixed-effects discrete-time, univariate SSM in which $\sigma_r = 0$ and $\lambda = 1$, which reduces to a first-order autoregression in family i, twin j:

$$y_{i,j,t} = \beta_{i,j} y_{i,j,t-1} + q_t, \quad q \sim \mathcal{N}(0, \sigma_q), \tag{20}$$

$$\beta_{i,j} \sim \mathcal{MVN}(\mu_\beta, \Sigma_\beta(ACE)), \tag{21}$$

The aim of the simulation was to compare accuracy and power between the two step and hierarchical approaches across different sample sizes and heritabilities. The simulation parameters were set to $\sigma_\beta = .1$, $\sigma_q = 1$, $\mu_\beta = .5$, $a^2 \in \{.4, .8\}$, $c^2 = 0$, and $e^2 \in \{.4, .2\}$. The heritabilities chosen were intended to first allow clear discernment of statistical properties of each approach in limited sample with a large effect ($a^2 = .8$), and then to verify those properties under a more conservative and realistic scenario ($a^2 = .4$).

Sample dimensions included $T, N \in \{50, 200\}$ occasions per twin and twin pairs per zygosity, making the design $2^4 = 16$ total conditions. β values were generated randomly from the twin covariance matrix, then individual time series were generated using each value of β. Power was defined as the probability of rejecting a null hypothesis that $a^2 = 0$. Each condition was simulated 100 times to produce empirical parameter distributions.

Table 1 contains the mean point estimates, inner 95% intervals, and approximate power calculations for each condition. When within-person samples were few ($T = 50$), the hierarchical model had 50% greater power to detect a non-zero heritability than the two-step analysis. The mean heritability estimates under each condition shows that as the number of families increased ($N = 200$ twin pairs per zygosity), the hierarchical model came closer to the data-generating heritabilities, while the 2-step analysis remained below

an upper boundary determined by the within-person sample size. With $N = 200, T = 50$, the average 2-step estimate was only a quarter of the true heritability.

Table 1
Heritability point estimates and power comparisons between 2-step maximum likelihood and hierarchical modeling with MCMC.

			Mean Estimate (95% CI)		Power	
			2-Step	Hier.	2-Step	Hier.
$a^2 = .8$	N=50	T=50	0.18 (-0.11, 0.47)	0.41 (-0.08, 0.9)	0.24	0.36
		T=200	0.46 (0.13, 0.79)	0.6 (0.21, 0.99)	0.79	0.86
	N=200	T=50	0.21 (0, 0.42)	0.57 (0.13, 1.01)	0.49	0.73
		T=200	0.53 (0.34, 0.72)	0.7 (0.46, 0.94)	>.99	>.99
$a^2 = .4$	N=50	T=50	0.09 (-0.14, 0.32)	0.27 (-0.1, 0.64)	0.12	0.29
		T=200	0.21 (-0.11, 0.53)	0.28 (-0.06, 0.62)	0.25	0.35
	N=200	T=50	0.1 (-0.05, 0.25)	0.27 (-0.05, 0.59)	0.27	0.38
		T=200	0.23 (0.02, 0.44)	0.3 (0.06, 0.54)	0.58	0.67

Figure 1 shows the distributions of point estimates for each iteration of simulation, with the 2-step estimates on the x-axes, and hierarchical model-based estimates on the y-axes. The minimum parameter standard error in the 2-step ML strategy results in a low, insurmountable boundary on the maximum value of a^2 that can be recovered, while estimates from the hierarchical model occasionally came close to the data-generating value.

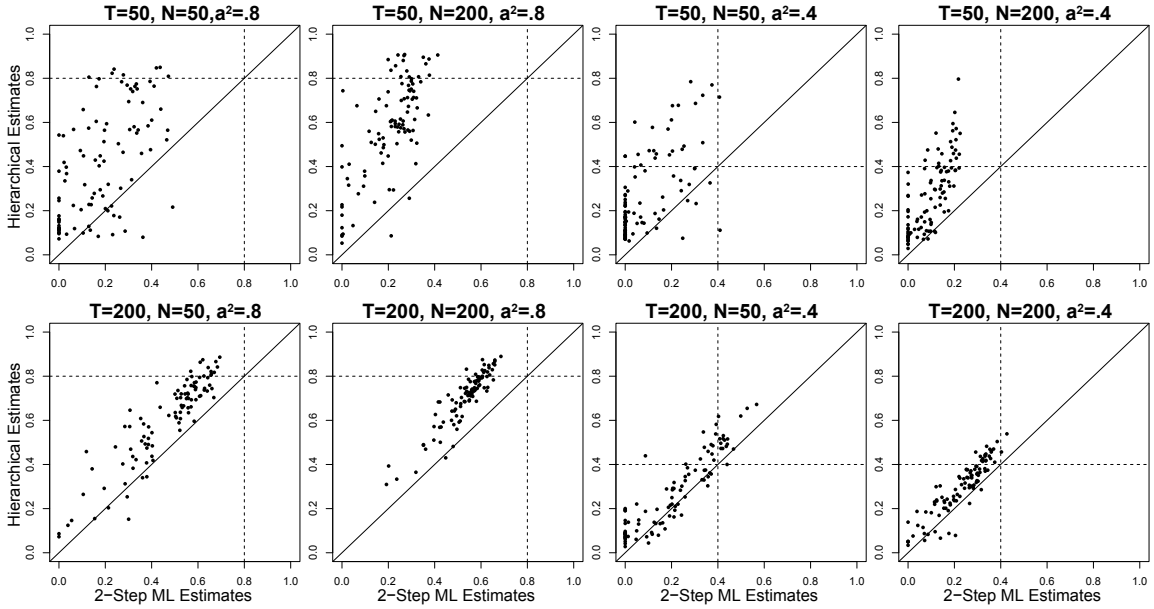

Figure 1. Pairwise comparison of heritability estimates from each method and condition over 100 trials of simulation. X-axes: 2-step approach; Y-axes: hierarchical approach. Dashed lines: true, data-generating heritabilities. All cases substantial downward bias in the 2-step method that is reduced in the hierarchical method.

Smaller T results in larger sampling variance of $\beta_{i,j}$, diluting its twin covariances and leading to downward bias of a^2. The hierarchical modeling approach greatly reduced the downward bias by incorporating shared statistical information across individuals into the individual parameter estimates. Conversely, smaller values of N primarily lead to a larger variances of a^2 in both models. As $T \to \infty$, the power to detect the heritability of the dynamics conditional on N in the 2-step ML analysis should approach that of a classical univariate twin model.

As a final note on power, these cursory simulations are only intended to demonstrate the dependence of bias, variance, and power on sample size. They are not to be taken as a power calculation table in general, as the precision of a dynamic parameter depends on the specification of the model, the signal-to-noise ratio, dependence between parameters, and empirical identification in certain nonlinear models. (e.g., McKee and Neale, 2019).

Data Application: Heart Rate Dynamics

Many time-varying phenotypes will warrant more complex models than lag-1 autoregressions. Take for instance the heart rate series shown in Figure 2, which exhibits gradual oscillations over time with random changes in amplitude. Second-order differential equations are suitable for data that show this kind of smooth random variation.

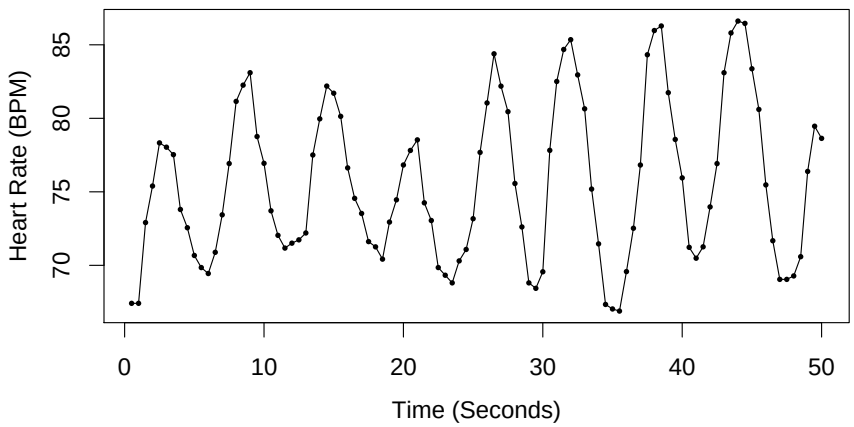

Figure 2. Example 50-second series of heart rate in beats-per-minute (BPM).

The following section demonstrates an application of hierarchical SSMs to real physiological data from which the above example was drawn. The model will describe a bivariate heritability structure of two parameters controlling the shape of heart rate series.

Data

EKG data were collected as part of a series of biometrical genetic studies of anxiety in 155 monozygotic and 240 dizygotic twin pairs conducted by Roberson-Nay et. al. at the Virginia Institute of Psychiatric and Behavioral Genetics (Roberson-Nay et al., 2013, 2014, 2012). The experiment involved a 10 minute pre-exposure period of physiological monitoring followed by administration of 7% CO_2 to induce elevated physiological arousal and gauge anxiety reactions for another 10 minutes. Samples of electrocardiogram (EKG) were transformed into beats-per-minute at 0.5-second intervals using the weighted averages

of spectrogram peaks over 60-second intervals. The total time series involved both the pre-exposure and exposure periods and exhibited a complex nonlinear dynamic structure. To isolate and model only short-term oscillations frequencies and to reduce computational burden for this demonstration, EKG data were subset from the first half of the experiment to 50-second series, starting from 100 seconds into the experiment as a warm-up or buffer period for consistency. Series were mean-centered and divided by their standard deviations. The measurement interval was one second per observation of heart rate. The sample dimensions were thus 240 families $\times$ 2 twins $\times$ 100 time points per twin. No data were missing from the sample.

Model

A suitable model to describe the random oscillatory behavior of heart rate seen in Figure 2 is the stochastic damped linear oscillator. This model is a continuous-time, second-order SDE. Under certain parameterizations, solutions of the ordinary differential equation will be the product of an exponential decay function and sinusoid, or in other words, oscillations that decay in amplitude over time. The stochastic version may be thought of as a damped oscillator undergoing small but continual perturbations that result in variable amplitudes. The intrinsic decay in amplitude may be continuously active but difficult to observe amidst the random disturbances.

For this model, we may be interested in the genetic and environmental sources of variation in two dynamics, the frequency of oscillation and the rate of decay. It can be specified as a continuous-time SSM with random effects. For family i, twin j,

$$\begin{bmatrix}\ddot{x}\\ \dot{x}\end{bmatrix}_{i,j}(t) = \begin{bmatrix}0 & 1\\ -\eta_{i,j}^2 & -2\zeta_{i,j}\sqrt{\eta_{i,j}}\end{bmatrix}\begin{bmatrix}x\\ \dot{x}\end{bmatrix}_{i,j}(t) + \begin{bmatrix}0\\ q\end{bmatrix}_{i,j}(t), \tag{22}$$

$$q_{i,j}(t) \sim \mathcal{N}(0, \sigma_{q,i,j}), \tag{23}$$

The measurement equation for the model simply accounts for the measurement error of the univariate series, which in this case may result from both the heart rate sensor and coarseness in conversion of raw EKG data to beats-per-minute. A person-specific mean parameter is also included, as the simple series mean is not necessarily the best centroid for the dynamics.

$$y_{i,j}(t) = x_{i,j}(t) + \mu_{i,j} + r_{i,j}(t), \tag{24}$$

$$r_{i,j}(t) \sim \mathcal{N}(0, \sigma_{r,i,j}) \tag{25}$$

The transition matrix in Eq. 22 has two parameters, η, the natural frequency, and ζ, the damping ratio. Together, they respectively describe the frequency of oscillations around a mean value and the rate at which the amplitude of oscillation declines on average. A zero value of ζ results in *undamped* oscillation, while a value of one produces *critical damping* in which oscillations cease and the system decays monotonically to its mean value. When the system is stochastic, these behaviors are considered on average, but the continual injection of Gaussian-distributed disturbances (q) provides that the system is unlikely to settle at its mean in practice.

No cross-twin disturbance covariance or cotwin influence effects were estimated in this model, as twins had no contact during the experiment.

As both parameters describe distinct, uncorrelated features of the series, we will estimate both the unique and shared sources of genetic and environmental variance. The bivariate genetic and environmental structure was specified as three distinct components: one unique to each phenotype and one representing the set of shared genetic and environmental sources among them. Using the identity matrix of dimension k, I_k,

$$\{\eta_{i,1}, \zeta_{i,1}, \eta_{i,2}, \zeta_{i,2}\} \sim \mathcal{MVN}([\eta_\mu, \zeta_\mu, \eta_\mu, \zeta_\mu], \Theta), \tag{26}$$

$$\Theta = \mathbf{JAJ}^\mathsf{T} + \mathbf{KCK}^\mathsf{T} + \mathbf{LEL}^\mathsf{T}, \tag{27}$$

$$\mathbf{J}, \mathbf{K}, \mathbf{L} = I_2 \otimes \begin{bmatrix} \theta_\eta & r_\theta & 0 \\ 0 & r_\theta & \theta_\zeta \end{bmatrix} \quad \mathbf{A} = \begin{bmatrix} I_3 & zI_3 \\ zI_3 & I_3 \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} I_3 & I_3 \\ I_3 & I_3 \end{bmatrix}, \quad \mathbf{E} = \begin{bmatrix} I_3 & 0 \\ 0 & I_3 \end{bmatrix}, \tag{28}$$

where θ and r_θ are for J, K and L respectively the factor loadings onto unique and shared genetic (a_η, r_a, a_ζ), common environmental (c_η, r_c, c_ζ), and unique environmental (e_η, r_e, e_ζ) factors of η and ζ. This parameterization of the bivariate variance components ensures that the final covariance matrix Θ is positive definite while retaining interpretation of θ_η and θ_ζ as the phenotype-specific variance contributions and r_θ as that which is shared between them. Totals for each genetic and environmental variance component were computed post-hoc as $\theta^2 + r_\theta^2$.

Results

All point estimates were taken as the multivariate MAP value from the sampling distribution. Confidence intervals were computed using the 95% quantiles of the samples.

The sample-wide mean of frequency was estimated at $\hat{\eta}_\mu = .91$ (.89, .92), with a group standard deviation of $\hat{\sigma}_\eta = .16$ (.15, .17). 95% of individual mean frequency values fell between .64 and 1.19. With period equal to $2\pi/\eta$, that comes out to 5.27 to 9.84 seconds per cycle. The sample-wide mean of damping was estimated at $\hat{\zeta}_\mu = .45$ (.43, .47) with a group standard deviation of $\hat{\sigma}_\zeta = .21$ (.20, 22). 95% of $\hat{\sigma}_{q,i,j}$ estimates fell between .2 and 3.26, while $\hat{\sigma}_r$ was in all cases close enough to zero to be negligible.

Table 2 gives the MAP estimates for the proportion of the total variance of random frequency and damping parameters contributed by genetic and environmental factors. $\hat{c}^2$ components were near zero for both phenotypes, and no genetic or environmental correlations were found. To obtain estimates for a submodel without a c^2 component, posterior samples were subset to only those where $\hat{c}^2$ was close to zero and the statistics were recomputed. Frequency was estimated to have a heritability of .32 (.01, .42), while the heritability of the damping ratio was estimate at .5, (.17, .65). Confidence intervals on the heritability estimates were smaller where $\hat{c}^2$ was closer to zero. Overall sampling variance was high for $\hat{a}^2$ and $\hat{c}^2$, but confidence intervals around $\hat{a}^2$ did not include zero for either phenotype under the full model with all three components.

Figure 3 shows the bivariate sampling distributions of the genetic and environmental variance components for the frequency and damping ratio parameters. The bivariate sampling distributions of $\hat{a}^2$ and $\hat{c}^2$ for both parameters reflect that while the MAP is at approximately zero, lower values of $\hat{a}^2$ are compensated by larger values of $\hat{c}^2$.

Table 2
Estimated proportions of total variance contributed by additive genetic (a^2), shared environmental (c^2), and unique environmental (e^2) factors, with 95% quantile confidence intervals and comparisons to a sub-model with $c^2 = 0$.

	ACE Model			AE Model ($c_\theta = 0$)		
	MAP	2.5%	97.5%	MAP	2.5%	97.5%
$\hat{a}^2_\eta$	0.32	(0.01,	0.42)	0.31	(0.16,	0.45)
$\hat{c}^2_\eta$	0.01	(0.00,	0.25)			
$\hat{e}^2_\eta$	0.67	(0.56,	0.85)	0.69	(0.54,	0.84)
$\hat{a}^2_\zeta$	0.50	(0.17,	0.65)	0.54	(0.37,	0.67)
$\hat{c}^2_\zeta$	0.01	(0.00,	0.20)			
$\hat{e}^2_\zeta$	0.49	(0.33,	0.70)	0.46	(0.32,	0.63)
$\hat{r}_a$	0.00	(0.00,	0.05)	0.00	(0.00,	0.03)
$\hat{r}_c$	0.03	(0.00,	0.59)			
$\hat{r}_e$	0.00	(0.00,	0.02)	0.00	(0.00,	0.02)

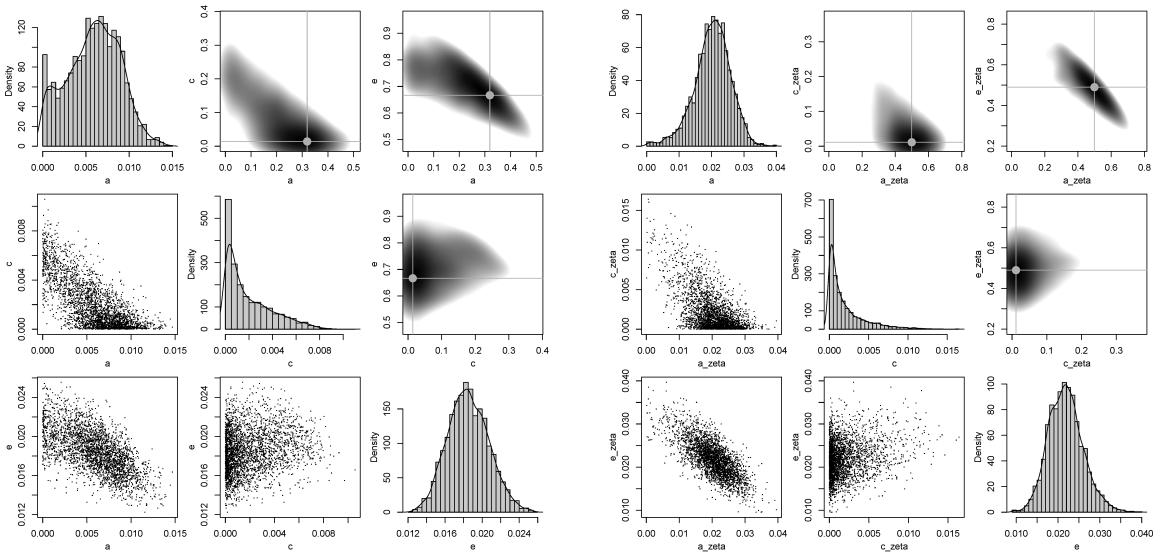

Figure 3. Diagonals: Marginal distributions; Lower triangle: Raw posterior samples; Upper triangle: Gaussian kernel densities of variance proportions with maxima. Sampling distribution for genetic ($\hat{a}^2$), common environmental $\hat{c}^2$, and unique environmental $\hat{e}^2$ variance components for frequency (left) and damping ratio (right).

Discussion

This study aimed to introduce and test the properties of hierarchical SSMs modeling for classical twin models in BG when one or more phenotypes are represented by the parameters of within-person models, namely SSMs. Hierarchical modeling is proposed to have substantial benefits over 2-step modeling, but requires more intensive computation. Simulations were used to test the statistical properties, and a data application to heart rate time series served as a reference for obtaining and interpreting parameter estimates.

The simulation compared accuracy and power of hierarchical heritability estimation. The two-step procedure suffers from insurmountable limits on the minimum amount of downward bias to the heritability estimate because the true twin correlations of the parameters are diluted by the parameters' sampling variance. Fitting a single-step hierarchical version of same model to iterations of simulation for two sample size conditions showed substantial increases in power and reductions in the expected downward bias, with many iterations coming close to the data-generating heritability value. The key finding of this simulation was that with the hierarchical model, accuracy and the power to detect the heritability of the phenotypic parameter continue to improve with larger number of participants even when each participant has only a small number of occasions, whereas within-person sample size limits the maximum detectible heritability of the 2-step analysis.

The application of a hierarchical SSM to heart rate series revealed moderate heritabilities of two heart rate dynamics: frequency of oscillation and damping ratio. No genetic or environmental variances were found to be shared between them. Estimates of the frequency and damping reveal about five to ten second cycles in heart rate that are expected to reduce in amplitude by about 45% per cycle on average when there are no further disturbances. The data were drawn from the initial, non-intervention half of an experiment in which CO_2 was administered to test anxiety sensitivity (e.g., Roberson-Nay et al., 2013). The results of this analysis may relate to cognitive aspects of anxiety, such as anticipation of the CO_2 intervention, but no covariates were included to determine whether anxiety sensitivity could explain the variation in the frequency and damping parameters. More elaborate, theory-driven models are warranted to test whether the genetic contributions to the dynamics found here relate to the underlying genetics of anxiety and panic.

Multivariate twin modeling is usually intended to investigate the shared genetic etiology of relatively well-established phenotypic constructs, such as DSM psychiatric disorders (Kendler et al., 2008; Tambs et al., 2009). With the methods provided here, it can also be used to obtain a deeper understanding of new phenotypes derived from theoretical person-level models. If two or more behavioral theories propose to explain the same time series data, then the joint multivariate genetic modeling of their parameters might allow insight into the extent to which the competing theories either jointly or uniquely represent biological pathways. Consider a hypothetical multivariate twin model that includes phenotypic dynamics from multiple, competing models of the heart rate fluctuations. A strong genetic covariance of parameters from different models would suggest that they may be representing similar biological pathways in different terms, whereas if genetic signals are unique to each model, then it may be the case that the competing models actually represent distinct and complementary mechanisms. Similarly, if the parameters of one particular model are heritable but do not share additive genetic sources of variation among themselves, that would seem to indicate that the model indexes multiple, distinct biological pathways. While the damped linear oscillator model of heart rate used in this analysis was not based on any particular theoretical background, the lack of correlation in the genetic signals of frequency and damping suggests that they could represent phenotypes with distinct biological structures.

Multivariate analyses of heritability is a unique tool for theoretical model validation because it enables us to link the ontology of new constructs together at any level of abstraction, even when the relevant biological paths cannot yet be known. This is potentially important for fields like cognitive science, where theories of mental processes are neces-

sarily posed in terms of emergent computational, statistical, and information-theoretic abstractions. The studies conducted here concerned similarly abstract cognitive phenotypes. Emotion regulation is likely a broad aggregate of cognitive and situational factors each contributing partial regulatory functions. It is likely too diffuse to relate to particular neural circuits, at least in the complex, *in situ* mode of EMA. To give one precedent, some components of posture control have been conceptualized in the cognitive domain as well. The tilt insensitivity radius estimated in McKee and Neale (2019) is thought to relate to the resolution of a cognitive model of one's body in space, and thus reliably estimating it would access an otherwise obscure fact of human perception and information processing.

Conclusion

Classical twin modeling and SSMs were combined in a single-step hierarchical modeling approach. Hierarchical models of intensive within-person data were shown to greatly increase power and accuracy of estimated biometrical genetic variance components over 2-step methods. A model suitable for experimental designs with physiological measures and independent trials between twins was fit to heart rate data, finding moderate, uncorrelated heritabilities of two heart rate dynamics. Hierarchical SSMs are thus an ideal strategy to study novel model-based phenotypes with intensive within-person data.

Acknowledgements

Special thanks go to Roxann Roberson-Nay (Virginia Commonwealth University) for providing the heart rate data and Daniel Lee (Columbia University) for technical troubleshooting, guidance with Stan model code. Additional thanks go to Michael Neale (Virginia Commonwealth University) and Steven Boker (University of Virginia) for prompting discussion and work on this topic.

References

Åström, K. J. and Murray, R. M. (2008). *Feedback systems : an introduction for scientists and engineers*. Princeton University Press, Princeton.

Bar-Shalom, Y., Li, X.-R., and Kirubarajan, T. (2002). *Estimation with Applications to Tracking and Navigation: Theory, Algorithms and Software*. John Wiley & Sons, Inc., New York, USA.

Bastos, A. and Schoffelen, J. (2016). A tutorial review of functional connectivity analysis methods and their interpretational pitfalls. *Frontiers in systems neuroscience*, 9:175–175.

Boker, S. M. (2002). Consequences of continuity: The hunt for intrinsic properties within parameters of dynamics in psychological processes. *Multivariate Behavioral Research*, 37(3):405–422.

Bollen, K. A. (1989). *Structural equations with latent variables*. Wiley series in probability and mathematical statistics. Applied probability and statistics. Wiley, New York. doi:10.1002/9781118619179.

Carpenter, B., Gelman, A., Hoffman, M. D., Lee, D., Goodrich, B., Betancourt, M., Brubaker, M., Guo, J., Li, P., and Riddell, A. (2017). Stan: A probabilistic programming language. *Journal of statistical software*, 76(1).

Dolan, C. V., de Kort, J. M., van Beijsterveldt, T. C. E. M., Bartels, M., and Boomsma, D. I. (2014). Ge covariance through phenotype to environment transmission: an assessment in longitudinal twin data and application to childhood anxiety. *Behavior genetics*, 44(3):240.

Douc, Randal, a. (2014). *Nonlinear times series : theory, methods and applications with R examples*. Texts in statistical science.

Driver, C. C., Oud, J. H. L., and Voelkle, M. C. (2017). Continuous time structural equation modeling with r package ctsem. *Journal of statistical software*, 77(5):1–35.

Eid, M. and Diener, E. (1999). Intraindividual variability in affect: Reliability, validity, and personality correlates. 76(4):662–676.

Eysenck, H. J. and Prell, D. B. (1951). The inheritance of neuroticism: an experimental study. *The Journal of mental science*, 97(408):441.

Hektner, J. M. (2007). *Experience sampling method : measuring the quality of everyday life*. Sage Publications, Thousand Oaks, Calif. doi:10.4135/9781412984201.

Hoffman, M. D. and Gelman, A. (2014). The no-u-turn sampler: adaptively setting path lengths in hamiltonian monte carlo. *J. Mach. Learn. Res.*, 15(1):1593–1623.

Hunter, M. D. (2018). State space modeling in an open source, modular, structural equation modeling environment. *Structural Equation Modeling: A Multidisciplinary Journal*, 25(2):307–324.

Kalman, R. E. and Bucy, R. S. (1961). New results in linear filtering and prediction theory. *Transactions of the ASME, Series D, Journal of Basic Engineering*, 83:95–108. doi:10.1115/1.3658902.

Kalman, R. E. et al. (1960). Contributions to the theory of optimal control. *Bol. soc. mat. mexicana*, 5(2):102–119.

Kendler, K. S., Aggen, S. H., Czajkowski, N., Røysamb, E., Tambs, K., Torgersen, S., Neale, M. C., and Reichborn-Kjennerud, T. (2008). The structure of genetic and environmental risk factors for dsm-iv personality disorders: a multivariate twin study. *Archives of General Psychiatry*, 65(12):1438–1446.

Kuppens, P., Mechelen, I. V., Nezlek, J. B., Dossche, D., and Timmermans, T. (2007). Individual differences in core affect variability and their relationship to personality and psychological adjustment. *Emotion (Washington, D.C.)*, 7(2):262–274.

MacKay, D. J. (2003). *Information theory, inference and learning algorithms*. Cambridge university press.

Mcardle, J. J. and Epstein, D. (1987). Latent growth curves within developmental structural equation models. *Child Development*, 58(1):110–133.

McElreath, R. (2016). *Statistical Rethinking: A Bayesian Course with Examples in R and Stan*, volume 122 of *Chapman & Hall/CRC Texts in Statistical Science*. CRC Press, Boca Raton, 1 edition.

McKee, K., Russell, M., Mennis, J., Mason, M., and Neale, M. (2020). Emotion regulation dynamics predict substance use in high-risk adolescents. *Addictive Behaviors*, page 106374.

McKee, K. L. and Neale, M. C. (2019). Direct estimation of the parameters of a delayed, intermittent activation feedback model of postural sway during quiet standing. *PloS one*, 14(9).

McNeish, D. and Hamaker, E. L. (2020). A primer on two-level dynamic structural equation models for intensive longitudinal data in mplus. *Psychological methods*, 25(5):610–635.

Moscati, A., Verhulst, B., McKee, K., Silberg, J., and Eaves, L. (2018). Cross-lagged analysis of interplay between differential traits in sibling pairs: Validation and application to parenting behavior and adhd symptomatology. *Behavior Genetics*, 48(1):22–33.

Neale, M. C., Hunter, M. D., Pritikin, J. N., Zahery, M., Brick, T. R., Kirkpatrick, R. M., Estabrook, R., Bates, T. C., Maes, H. H., and Boker, S. M. (2016). Openmx 2.0: Extended structural equation and statistical modeling. *Psychometrika*, 81(2):535.

Nesselroade, J. R. (2002). Elaborating the differential in differential psychology. *Multivariate Behavioral Research*, 37(4):543–561.

Oravecz, Z., Tuerlinckx, F., and Vandekerckhove, J. (2011). A hierarchical latent stochastic differential equation model for affective dynamics. *Psychological Methods*, 16(4):468–490. doi:10.1037/a0024375.

Ou, L., Hunter, Michael, D., and Chow, S.-M. (2019). Whats for dynr: A package for linear and nonlinear dynamic modeling in r. *The R journal*, 11(1):91.

Plomin, Robert, . (1990). *Nature and nurture : an introduction to human behavioral genetics*. Brooks/Cole Pub. Co., Pacific Grove, Ca.

R Core Team (2019). *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing, Vienna, Austria.

Rajendra Acharya, U., Paul Joseph, K., Kannathal, N., Lim, C. M., and Suri, J. S. (2006). Heart rate variability: a review. 44(12):1031–1051.

Roberson-Nay, R., Beadel, J. R., Gorlin, E. I., Latendresse, S. J., and Teachman, B. A. (2014). Examining the latent class structure of co2 hypersensitivity using time course trajectories of panic response systems. *Journal of Behavior Therapy and Experimental Psychiatry*, 47:68–76.

Roberson-Nay, R., Latendresse, S. J., and Kendler, K. S. (2012). A latent class approach to the external validation of respiratory and non-respiratory panic subtypes. *Psychological medicine*, 42(3):461–474.

Roberson-Nay, R., Moruzzi, S., Ogliari, A., Pezzica, E., Tambs, K., Kendler, K. S., and Battaglia, M. (2013). Evidence for distinct genetic effects associated with response to 35co2: Research article: Evidence for distinct genetic effects. *Depression and anxiety*, 30(3):259–266.

Röcke, C., Li, S.-C., and Smith, J. (2009). Intraindividual variability in positive and negative affect over 45 days: Do older adults fluctuate less than young adults? *Psychology and aging*, 24(4):863–878.

Shiffman, S., Stone, A. A., and Hufford, M. R. (2008). Ecological momentary assessment. *Annual review of clinical psychology*, 4:1. doi:10.1146/annurev.clinpsy.3.022806.091415.

Tambs, K., Czajkowsky, N., Neale, M. C., Reichborn-Kjennerud, T., Aggen, S. H., Harris, J. R., Kendler, K. S., et al. (2009). Structure of genetic and environmental risk factors for dimensional representations of dsm–iv anxiety disorders. *The British Journal of Psychiatry*, 195(4):301–307.

Vallacher, R. R., Van Geert, P., and Nowak, A. (2015). The intrinsic dynamics of psychological process. *Current Directions in Psychological Science*, 24(1):58–64.

Wei, W. W. S. (2006). *Time series analysis : univariate and multivariate methods*. Pearson Addison Wesley, Boston, 2nd ed.. edition. doi:10.1093/oxfordhb/9780199934898.013.0022.